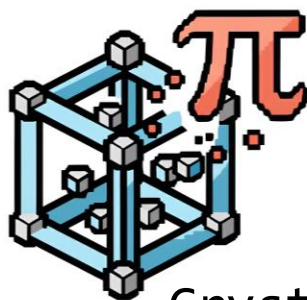
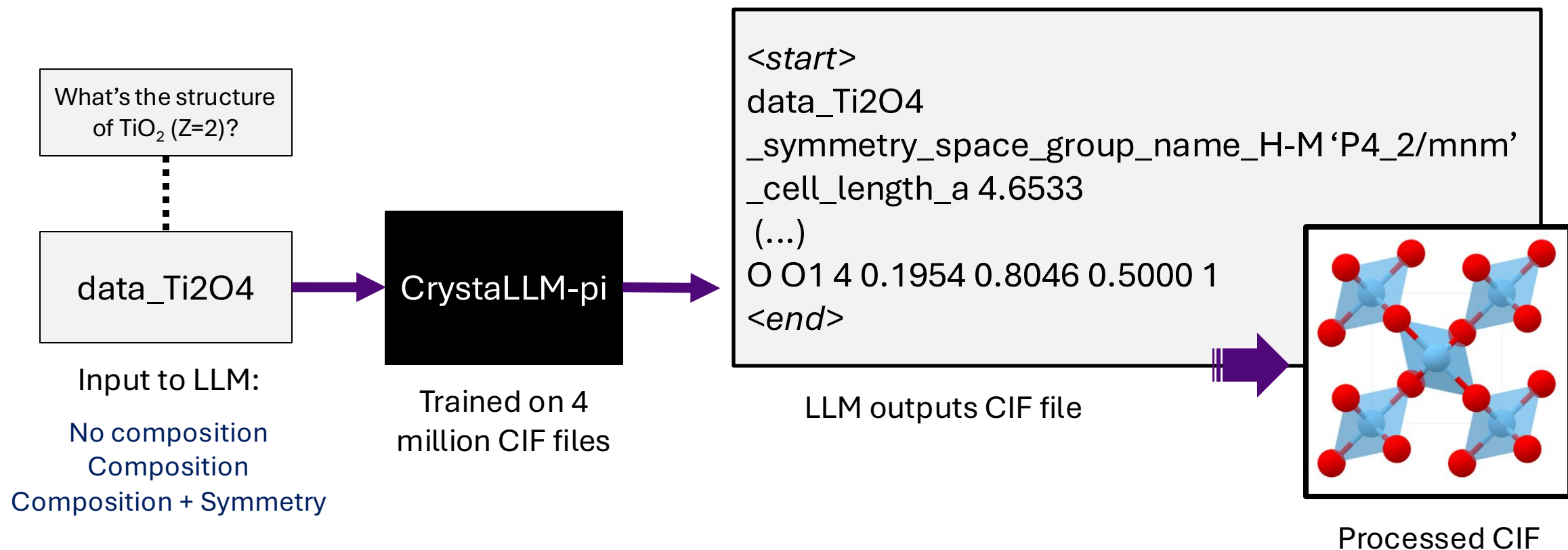


How to Condition Generative Transformers for Inorganic Crystals on Functional Properties



CrystaLLM- π



Generation of a Crystallographic Information File (CIF) using a transformer

2 Objectives:

1. Transfer knowledge from the unconditioned pre-training phase (lots of data) to a conditional fine-tuning phase (data usually scarce)
2. We want to condition outputs strongly on target properties

Raw Baseline: Append Properties as Text

Training data

```
<start>
_PBE_bandgap_eV 1.781
_stability_eV_atom 0.044
data_Ti2O4
_symmetry_space_group_name_H-M 'P4_2/mnm'
_cell_length_a 4.6533
(...)
O O1 4 0.195 4 0.8046 0.5000 1
<end>
```

*No model changes,
Easy to add to data*

*Digits tokenized
separately*

0.044

*Same token
embeddings for
structure + property*

4 : stability
4 : cell_length

Problematic when fine-tuning data is scarce...

A Better Representation for Properties

Property Labels

[1.781, 0.044]

Condition Encoder

Learned Property
Embedding

Unchanged CIF

```
<start>
data_Ti2O4
_symmetry_space_group_name_H-M 'P4_2/mnm'
_cell_length_a 4.6533
(...)
O O1 4 0.1954 0.8046 0.5000 1
<end>
```

*Continuous,
richer and data-
efficient property
representation*

*Requires learning a
separate property
encoder*

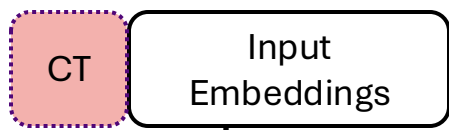
How should we inject this embedding into the transformer?

Where Should the Property Embedding Enter?

Learned Property Embedding

Condition Token (CT)

Prepended to CIF sequence



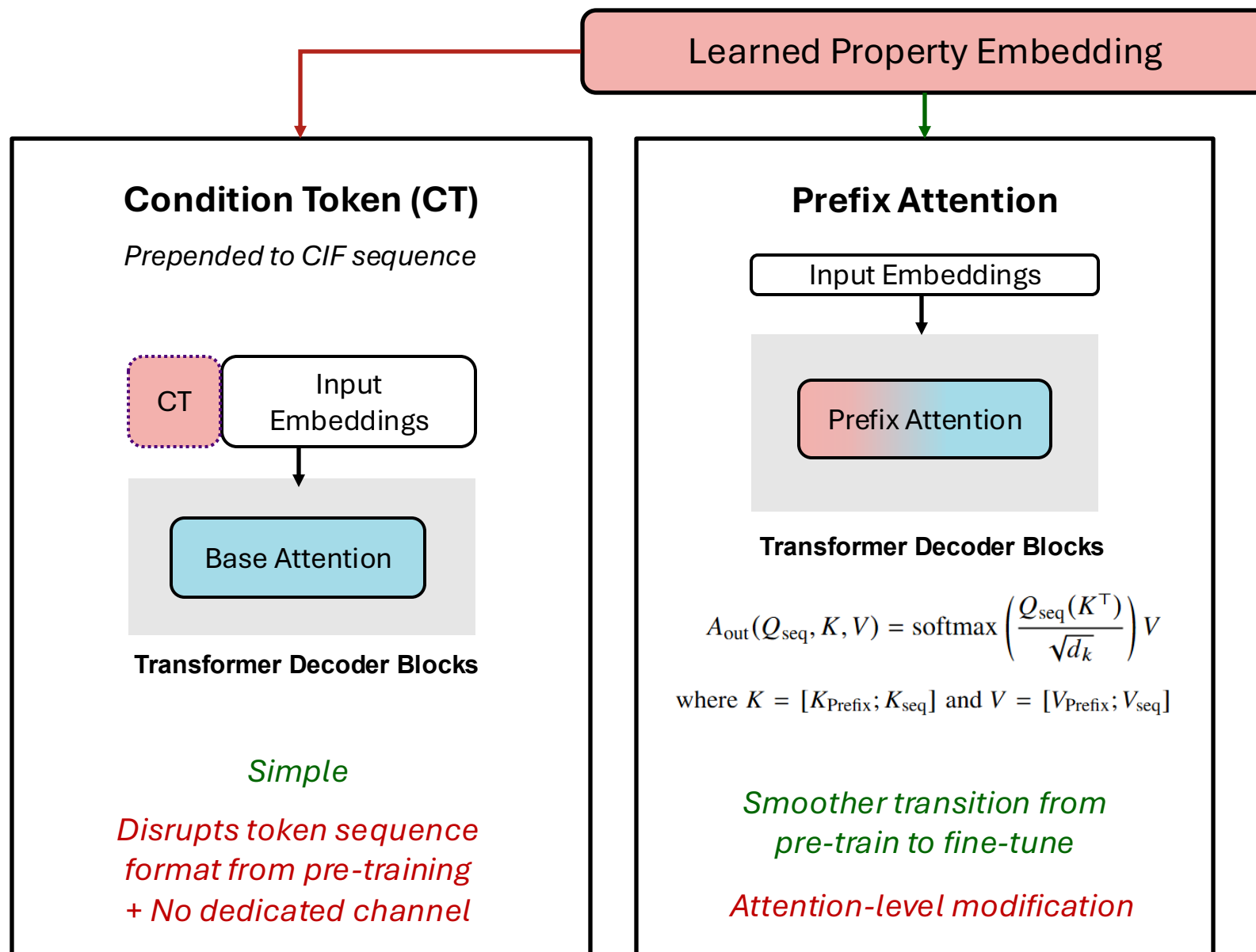
Base Attention

Transformer Decoder Blocks

Simple

*Disrupts token sequence
format from pre-training
+ No dedicated channel*

Where Should the Property Embedding Enter?

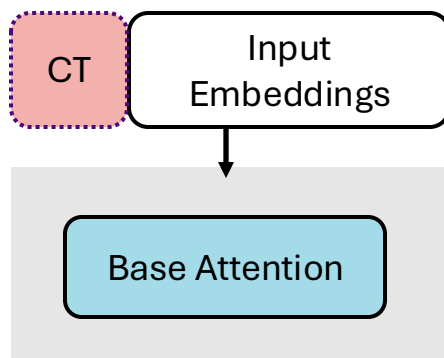


Where Should the Property Embedding Enter?

Learned Property Embedding

Condition Token (CT)

Prepended to CIF sequence



Transformer Decoder Blocks

Simple

*Disrupts token sequence
format from pre-training
+ No dedicated channel*

Prefix Attention

Input Embeddings



Transformer Decoder Blocks

$$A_{\text{out}}(Q_{\text{seq}}, K, V) = \text{softmax}\left(\frac{Q_{\text{seq}}(K^T)}{\sqrt{d_k}}\right)V$$

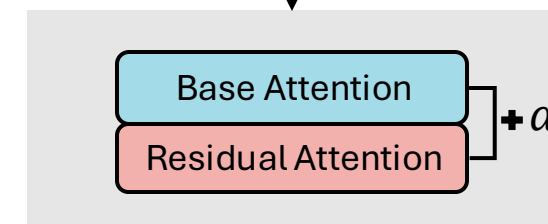
where $K = [K_{\text{Prefix}}; K_{\text{seq}}]$ and $V = [V_{\text{Prefix}}; V_{\text{seq}}]$

*Smoother transition from
pre-train to fine-tune*

Attention-level modification

Residual Attention

Input Embeddings



Transformer Decoder Blocks

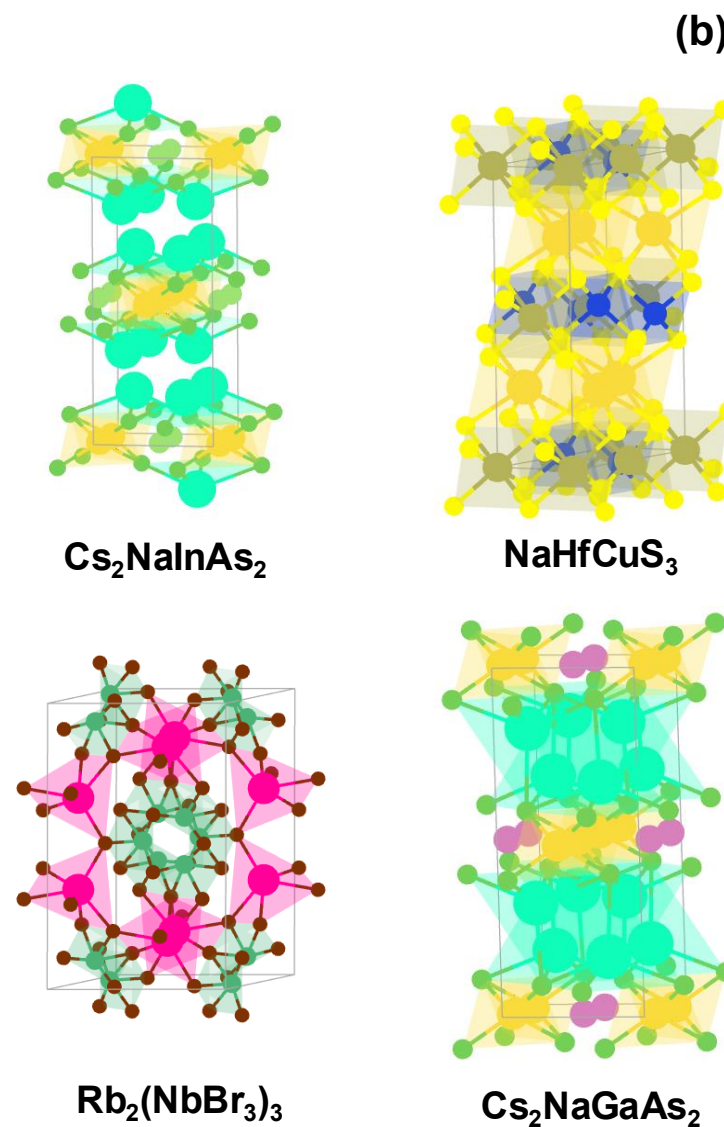
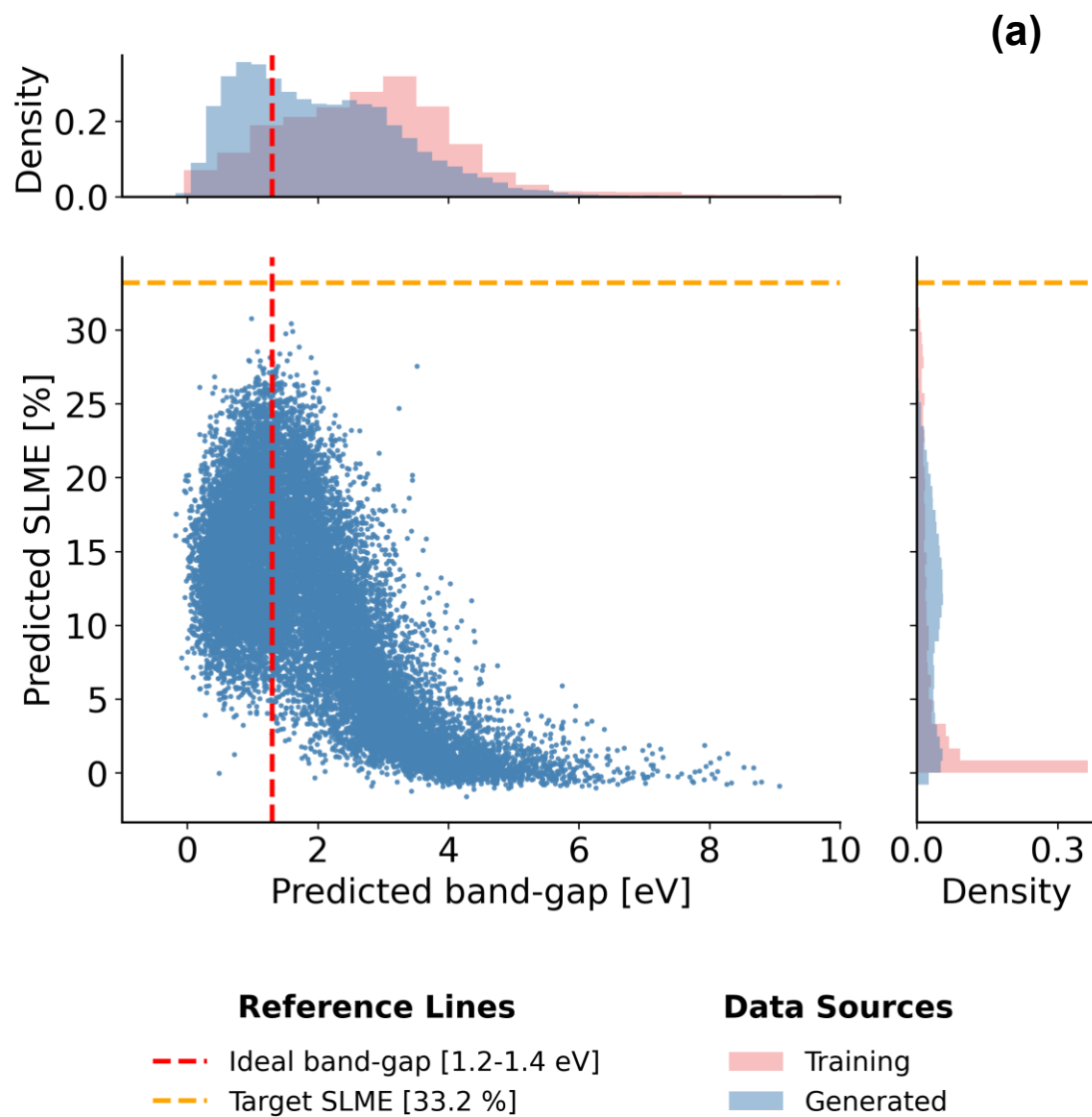
$$A_{\text{base}} = \text{softmax}\left(\frac{Q_{\text{seq}}K_{\text{seq}}^T}{\sqrt{d_k}}\right)V_{\text{seq}}$$

$$A_{\text{Residual}} = \text{softmax}\left(\frac{Q_{\text{seq}}K_{\text{Residual}}^T}{\sqrt{d_k}}\right)V_{\text{Residual}}$$

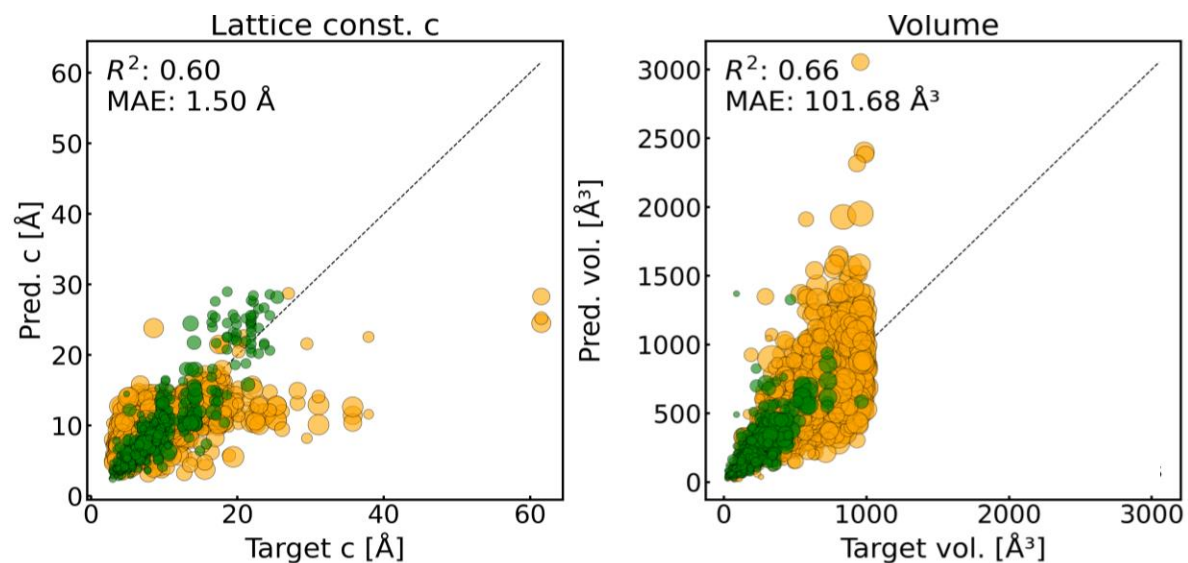
$$A_{\text{out}} = A_{\text{base}} + \alpha \cdot A_{\text{Residual}}$$

*Least Invasive + Handles
Missing Properties*

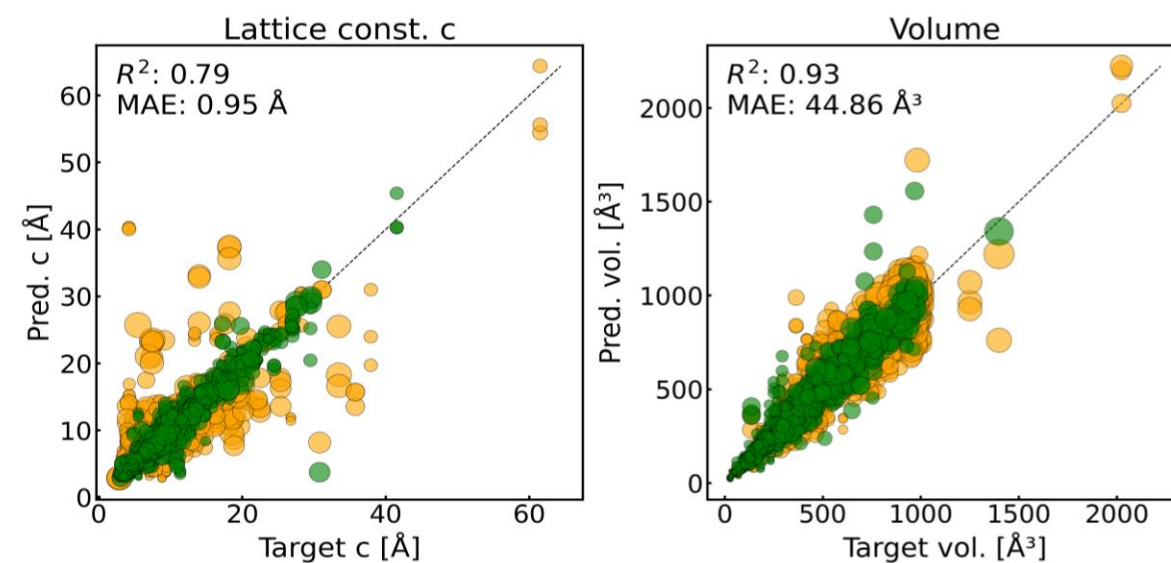
Harder to implement



b) No XRD Info



a) XRD Info



Structure Size

